



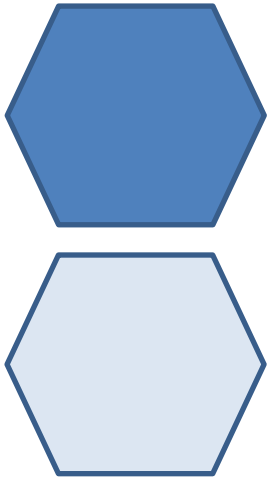
Bioinformatics

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Department of Computer, Control, and Management Engineering
Sapienza University of Rome

 Human Protein Atlas



The Human Protein Atlas

provides expression of human proteome and transcriptome
in all major tissues and organs

<https://www.proteinatlas.org/>

THE HUMAN PROTEIN ATLAS

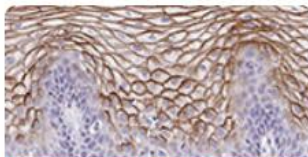
≡ MENU HELP NEWS

SEARCHⁱ

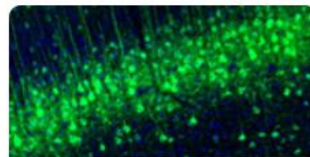
Search

[Fields »](#)

e.g. ACE2, GFAP, EGFR



TISSUE



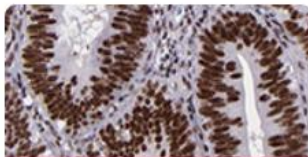
BRAIN



SINGLE CELL TYPE

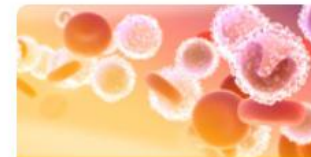


TISSUE CELL TYPE



PATHOLOGY

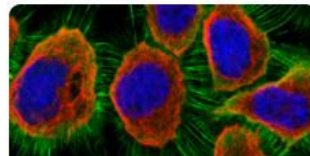
The open access resource for human proteins
Search for specific genes/proteins or
explore the 10 different sections



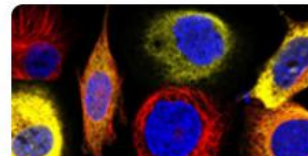
IMMUNE CELL



BLOOD PROTEIN



SUBCELLULAR



CELL LINE



METABOLIC

Example

THE HUMAN PROTEIN ATLAS

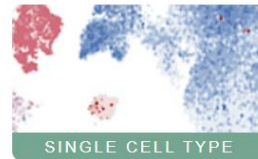
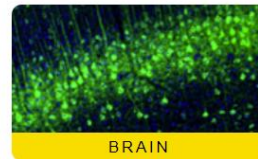
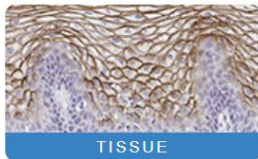
MENU HELP NEWS

SEARCHⁱ

TP53

e.g. ACE2, GFAP, EGFR

Search



46 GENES FOUNDⁱ

Show / hide columns ▼

Download: XML | RDF | TSV | JSON | Custom TSV/JSON ▼

Page 1 of 1

Gene ⁱ	Gene description ⁱ	x Evidence ⁱ x	Tissue ⁱ	Brain ⁱ	Single cell ⁱ	Tissue cell ⁱ	Pathology ⁱ	Immune ⁱ	Blood ⁱ	Subcell ⁱ	Cell line ⁱ	Metabolic ⁱ
TP53	Tumor protein p53	■		RNA	RNA	RNA		RNA	PEA		RNA	N/A
TP53INP1	Tumor protein p53 inducible nuclear protein 1	■		RNA	RNA	RNA		RNA	PEA		RNA	N/A

- Summary

THE HUMAN PROTEIN ATLAS

MENU

HELP

NEWS

TP53

Search

Fields »

Search result (46 genes): TP53 | TP53INP1 | TP53TG3 ...

TP53

SUMMARY

TISSUE

RNA BRAIN

RNA SINGLE CELL

RNA TISSUE CELL

PATHOLOGY

RNA IMMUNE

PEA BLOOD

RNA SUBCELL

RNA CELL LINE

N/A METABOLIC

PROTEIN SUMMARY

RNA DATA

GENE/PROTEIN

ANTIBODIES AND VALIDATION

Y

HUMAN PROTEIN ATLAS SUMMARYⁱ

Protein ⁱ	Tumor protein p53
Gene name ⁱ	TP53 (LFS1, p53)
Tissue specificity ⁱ	Low tissue specificity
Tissue expression cluster ⁱ	T-cells - Adaptive immune response (mainly)
Single cell type specificity ⁱ	Low cell type specificity
Single cell type expression cluster ⁱ	Non-specific - Translation (mainly)
Immune cell specificity ⁱ	Low immune cell specificity
Brain specificity ⁱ	Low human brain regional specificity
Cancer prognostic summary	Prognostic marker in endometrial cancer (favorable) and prostate cancer (unfavorable)
Predicted location ⁱ	Intracellular
Subcellular summary ⁱ	Located in Nucleoplasm, Mitochondria, Cytosol (Single cell variability, CCD Protein)
Protein function (UniProt) ⁱ	Acts as a tumor suppressor in many tumor types; induces growth arrest or apoptosis depending on the physiological circumstances and cell type. Involved in cell cycle regulation as a trans-activator that acts to negatively regulate cell division by controlling a set of genes required for this process. One of the activated genes is an inhibitor of cyclin-dependent kinases. Apoptosis induction seems to be mediated either by stimulation of BAX and FAS antigen expression, or by repression of Bcl-2 expression. Its pro-apoptotic activity is activated via its interaction with PPP1R13B/ASPP1 or TP53BP2/ASPP2 ¹ . However, this activity is inhibited when the interaction with PPP1R13B/ASPP1 or TP53BP2/ASPP2 is displaced by PPP1R13L/ASPP ² . In cooperation with mitochondrial PPIF is involved in activating oxidative stress-induced necrosis; the function is largely independent of transcription. Induces the transcription of long intergenic non-coding RNA p21 (lincRNA-p21) and lincRNA-Mkn1. LincRNA-p21 participates in TP53-dependent transcriptional repression leading to apoptosis and seems to have an effect on cell-cycle regulation. Implicated in Notch signaling cross-over. Prevents CDK7 kinase activity when associated to CAK complex in response to DNA damage, thus stopping cell cycle progression. Isoform 2 enhances the transactivation activity of isoform 1 from some but not all TP53-inducible promoters. Isoform 4 suppresses transactivation activity and impairs growth suppression mediated by isoform 1. Isoform 7 inhibits isoform 1-mediated apoptosis. Regulates the circadian clock by repressing CLOCK-ARNTL/BMAL1-mediated transcriptional activation of PER2 ³ . <i>show less</i>
Molecular function (UniProt) ⁱ	Activator, DNA-binding, Repressor

- Expression in **normal tissues**

THE HUMAN PROTEIN ATLAS

MENU HELP NEWS

TP53

Search result (46 genes): **TP53** | TP53INP1 | TP53TG3 ...

SUMMARY TISSUE BRAIN SINGLE CELL TISSUE CELL PATHOLOGY IMMUNE BLOOD SUBCELL CELL LINE N/A METABOLIC

TISSUE

PRIMARY DATA

TISSUES

GENE/PROTEIN

ANTIBODIES AND VALIDATION

Dictionary

Tissue proteome

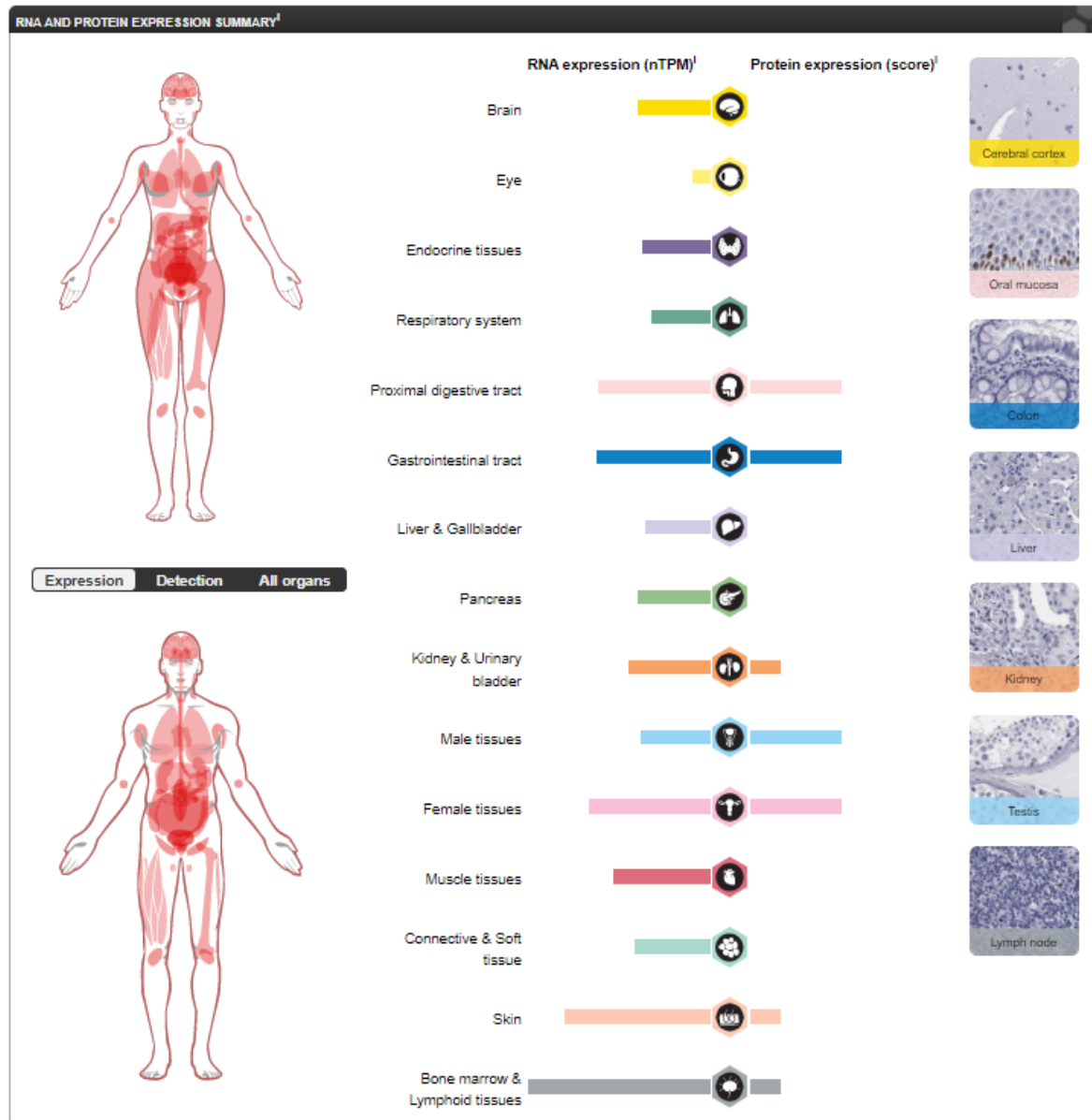
GENERAL INFORMATION ⁱ	
Gene name ⁱ	TP53
Gene description ⁱ	Tumor protein p53
Protein class ⁱ	Cancer-related genes Disease related genes Human disease related genes Plasma proteins Potential drug targets Transcription factors Transporters
Predicted location ⁱ	Intracellular
Number of transcripts ⁱ	23

HUMAN PROTEIN ATLAS INFORMATION ⁱ	
Tissue expression cluster (RNA) ⁱ	T-cells - Adaptive immune response (mainly)
Tissue specificity (RNA) ⁱ	Low tissue specificity
Tissue distribution (RNA) ⁱ	Detected in all
Protein evidence ⁱ	Evidence at protein level
Protein expression ⁱ	Fractions of cells showed weak nuclear and/or cytoplasmic expression.

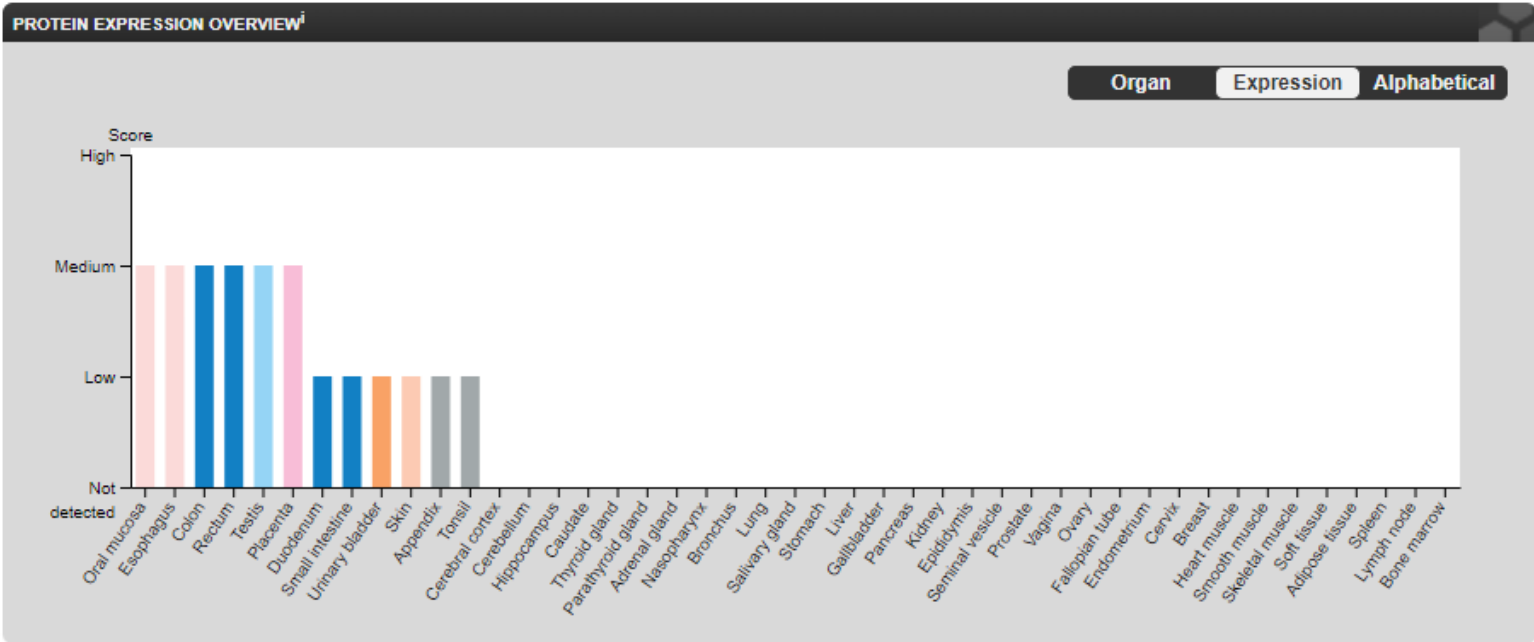
IMMUNOHISTOCHEMISTRY DATA RELIABILITY	
Data reliability description ⁱ	Medium consistency between antibody staining and RNA expression data. Tissue location of RNA and protein might differ due to short half-life of wild-type p53 protein.
Reliability score ⁱ	Enhanced
Antibodies ⁱ	CAB002973 , CAB039238 , CAB039239 , CAB072876

SHOW MORE

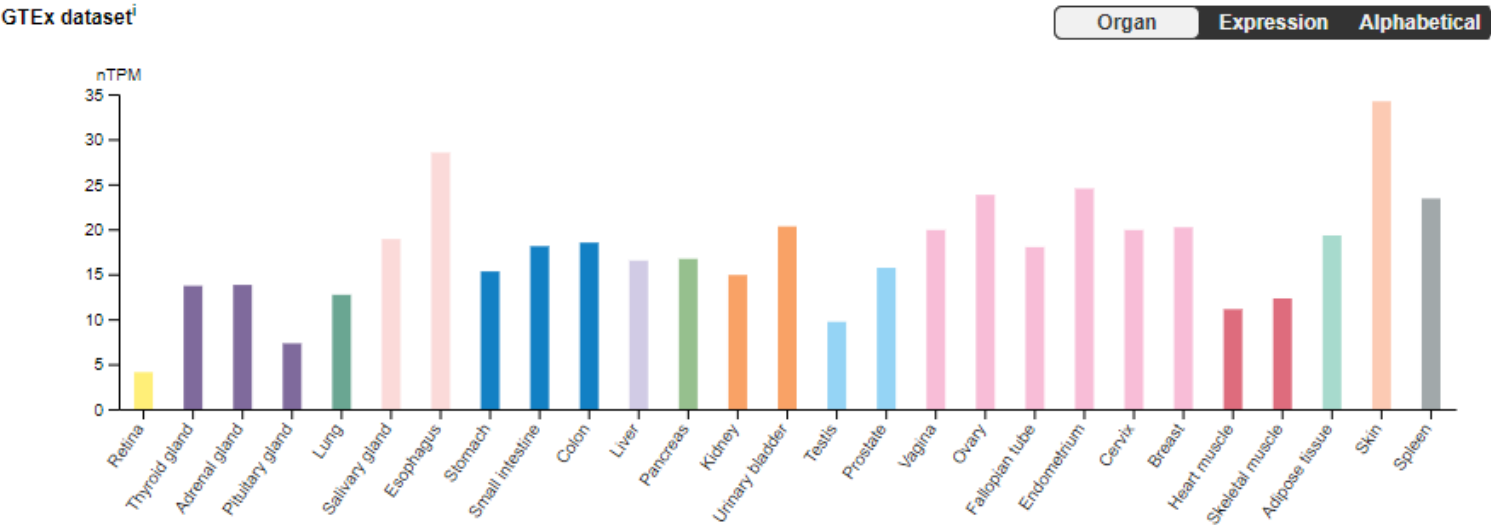
- Expression in **normal tissues**



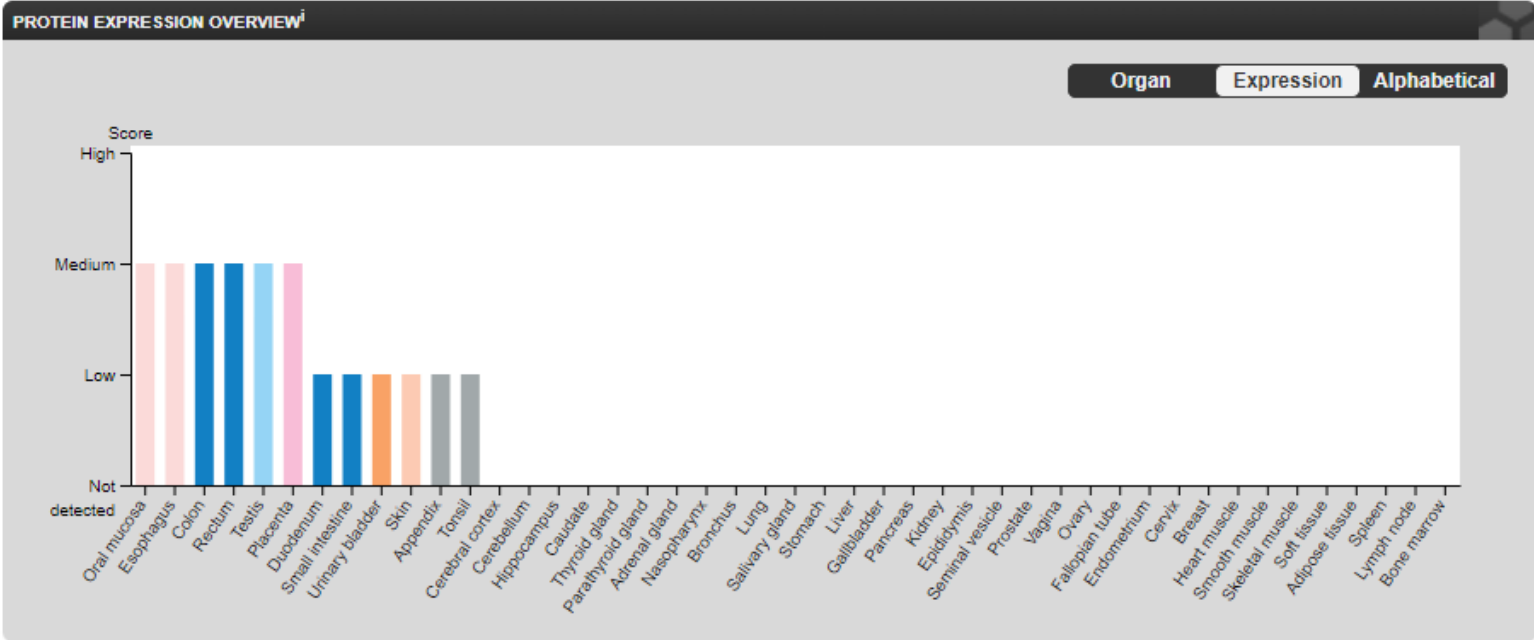
Protein expression



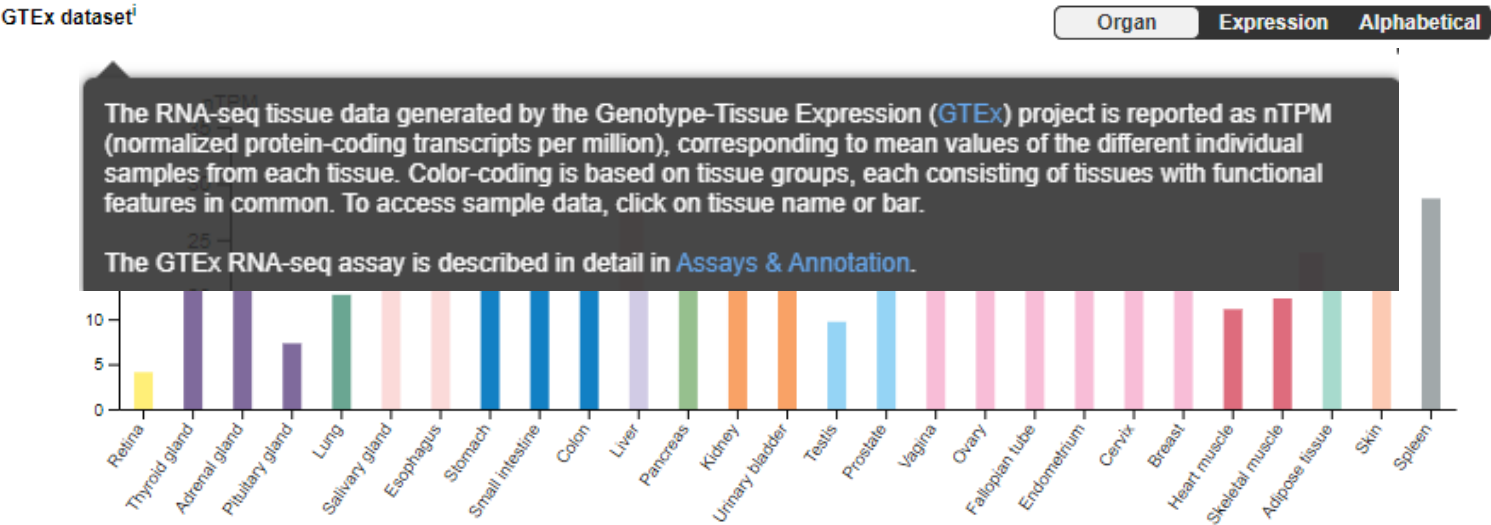
RNA



Protein expression



RNA



- Expression in **cancer tissues**

THE HUMAN PROTEIN ATLAS

TP53

Search result (48 genes): **TP53** | TP53INP1 | TP53TG3 ...

TP53

SUMMARY TISSUE BRAIN SINGLE CELL TISSUE CELL PATHOLOGY IMMUNE BLOOD SUBCELL CELL LINE METABOLIC

PATHOLOGY

CANCER

GENE/PROTEIN

ANTIBODIES AND VALIDATION

Dictionary

Human pathology

GENERAL INFORMATION¹

Gene name ¹	TP53
Gene description ¹	Tumor protein p53
Protein class ¹	Cancer-related genes Disease related genes Human disease related genes Plasma proteins Potential drug targets Transcription factors Transporters
Predicted location ¹	Intracellular
Number of transcripts ¹	23

HUMAN PROTEIN ATLAS INFORMATION¹

RNA category ¹	Consensus (human tissue): Detected in all
HPA (cell line):	Low cell line specificity Detected in all
TCGA (cancer tissue):	Low cancer specificity Detected in all
Protein evidence ¹	Evidence at protein level
Protein expression normal tissue ¹	Fractions of cells showed weak nuclear and/or cytoplasmic expression.

IMMUNOHISTOCHEMISTRY DATA RELIABILITY

Data reliability description ¹	Medium consistency between antibody staining and RNA expression data. Tissue location of RNA and protein might differ due to short half-life of wild-type p53 protein.
Reliability score - normal tissues ¹	Enhanced
Antibodies ¹	CAB002973 , CAB039238 , CAB039239 , CAB072876

SHOW MORE

PROGNOSTIC SUMMARY¹

Prognostic marker in **endometrial cancer** (favorable) and **prostate cancer** (unfavorable)

Endometrial cancer

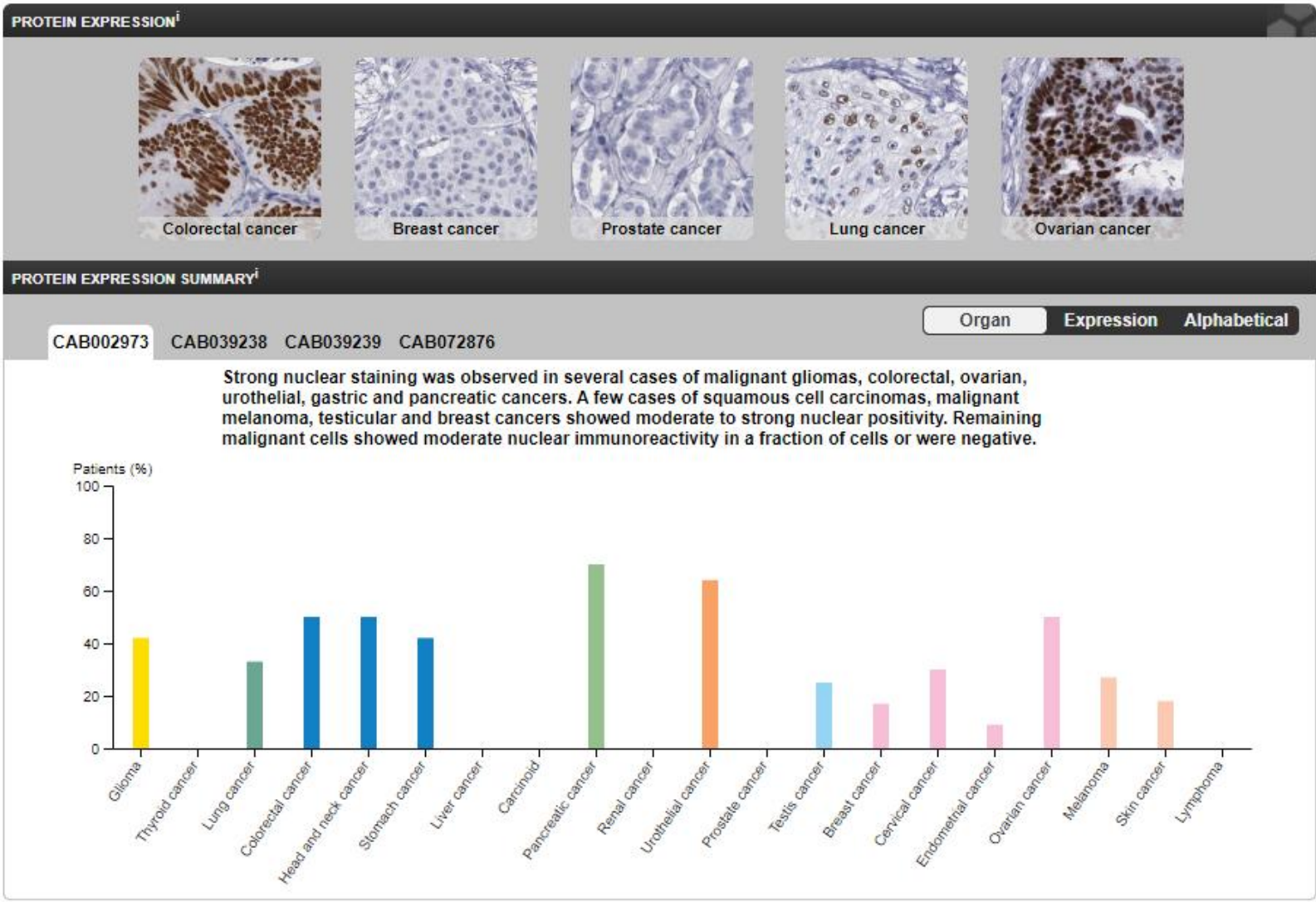
$p < 0.001$

Prostate cancer

$p < 0.001$

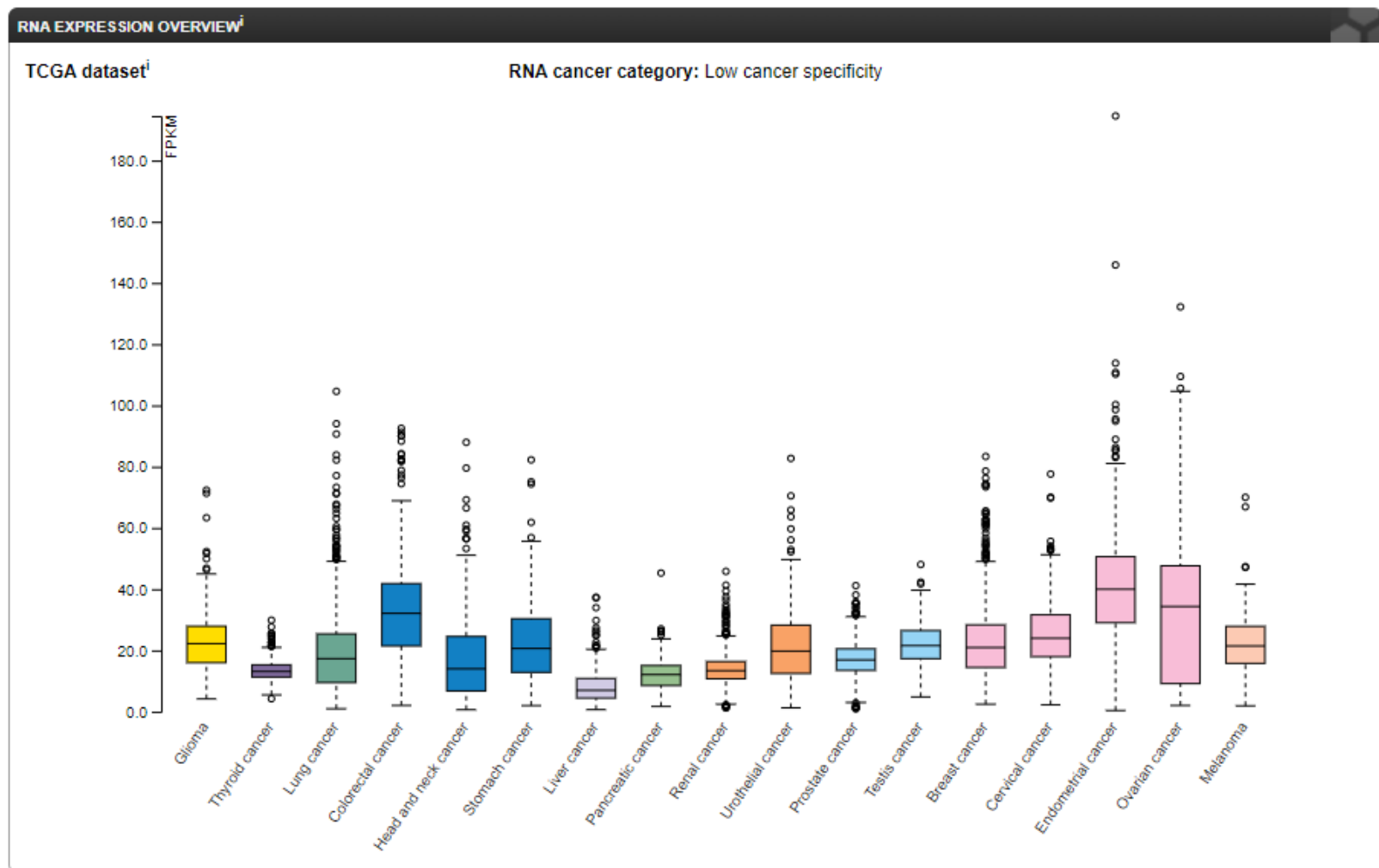
Protein expression

- Expression in **cancer tissues**



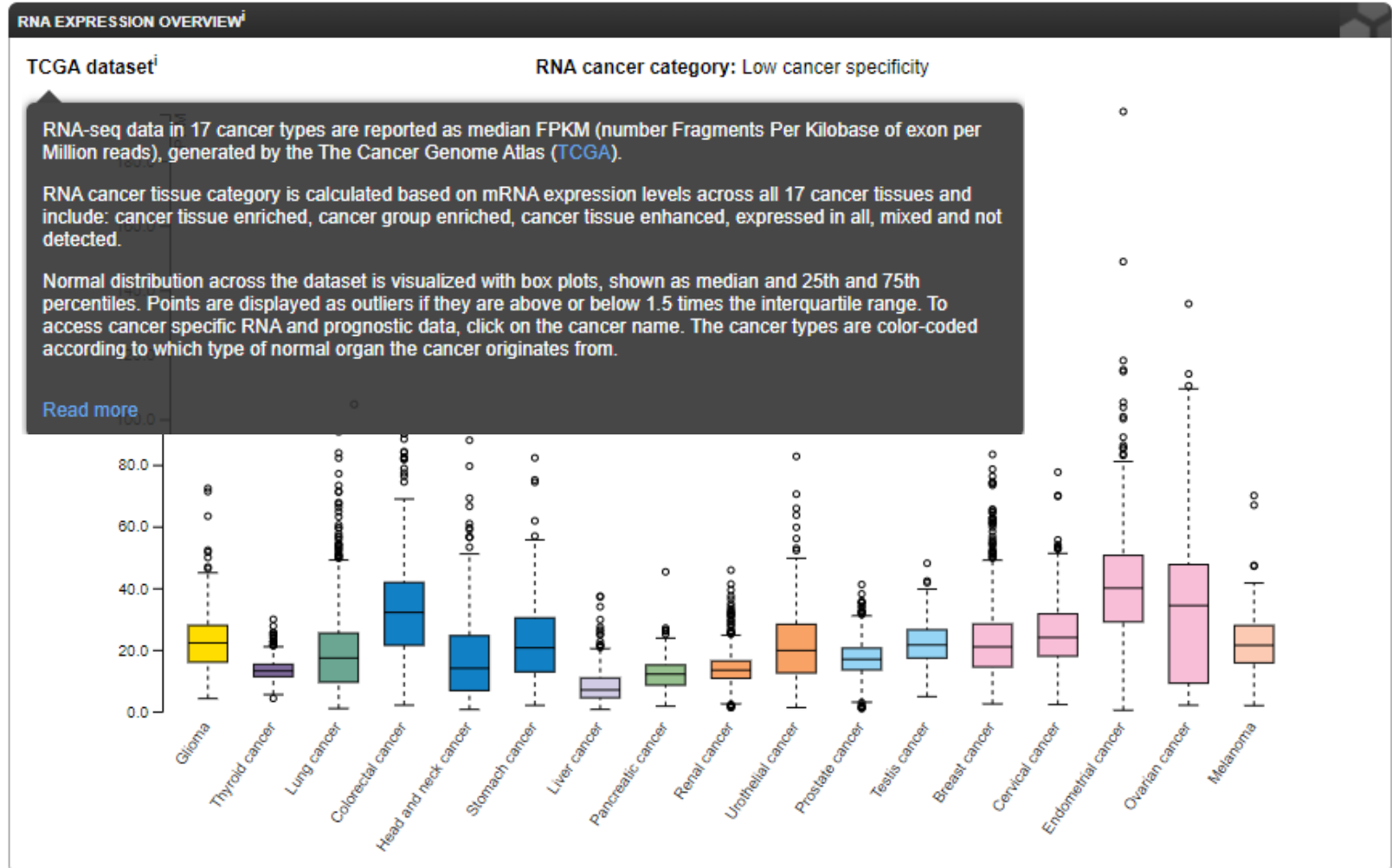
RNA expression

- Expression in **cancer tissues**



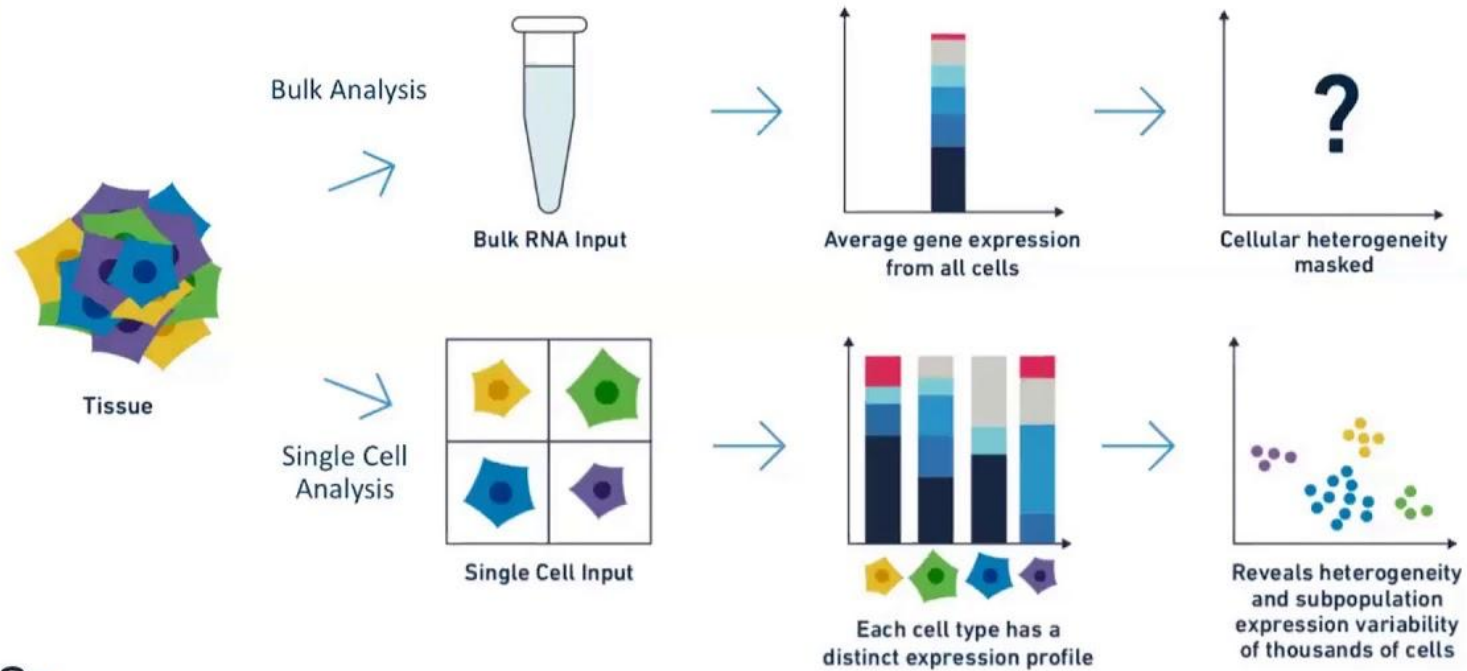
RNA expression

- Expression in **cancer tissues**

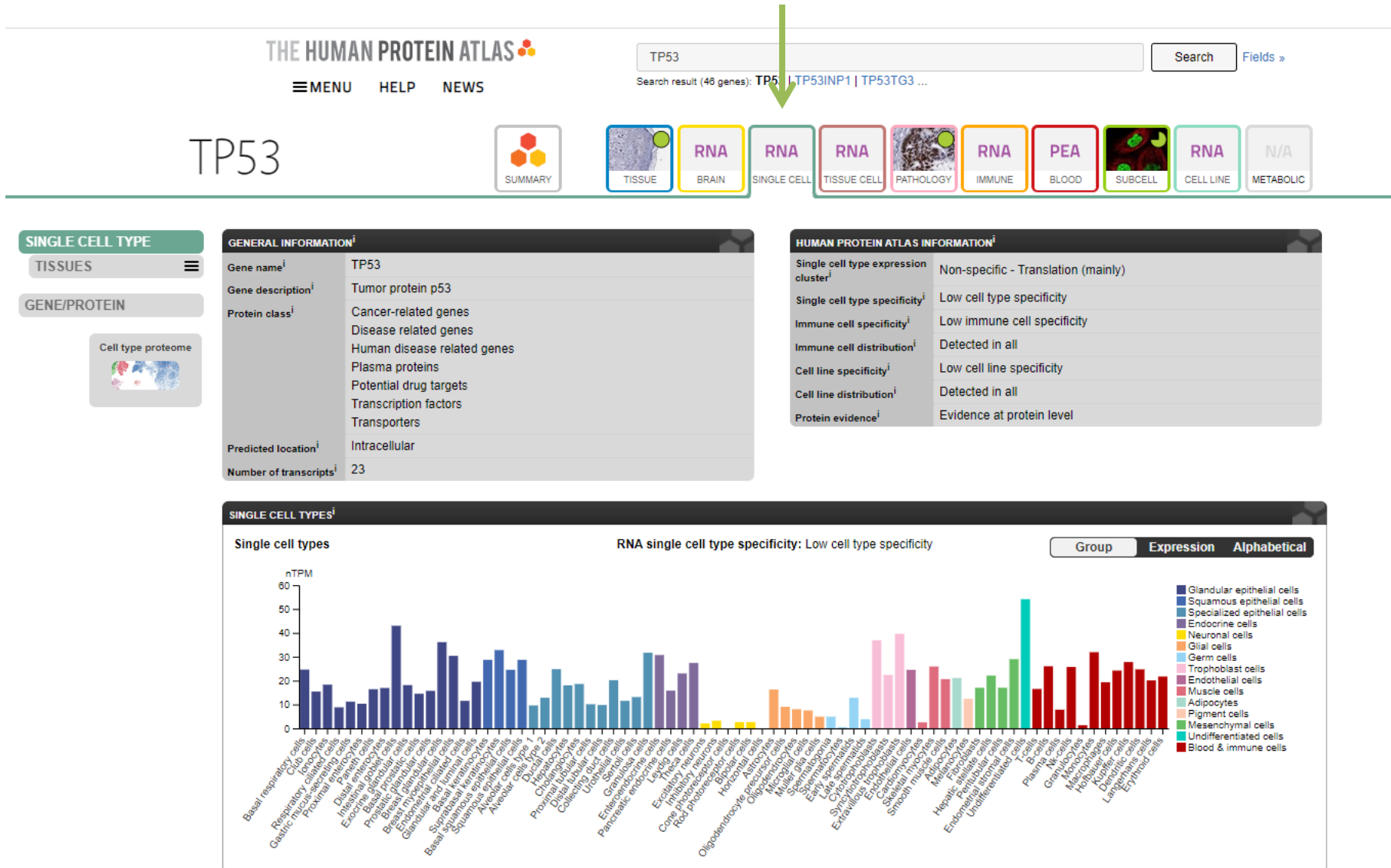


- Expression in **single cells population**

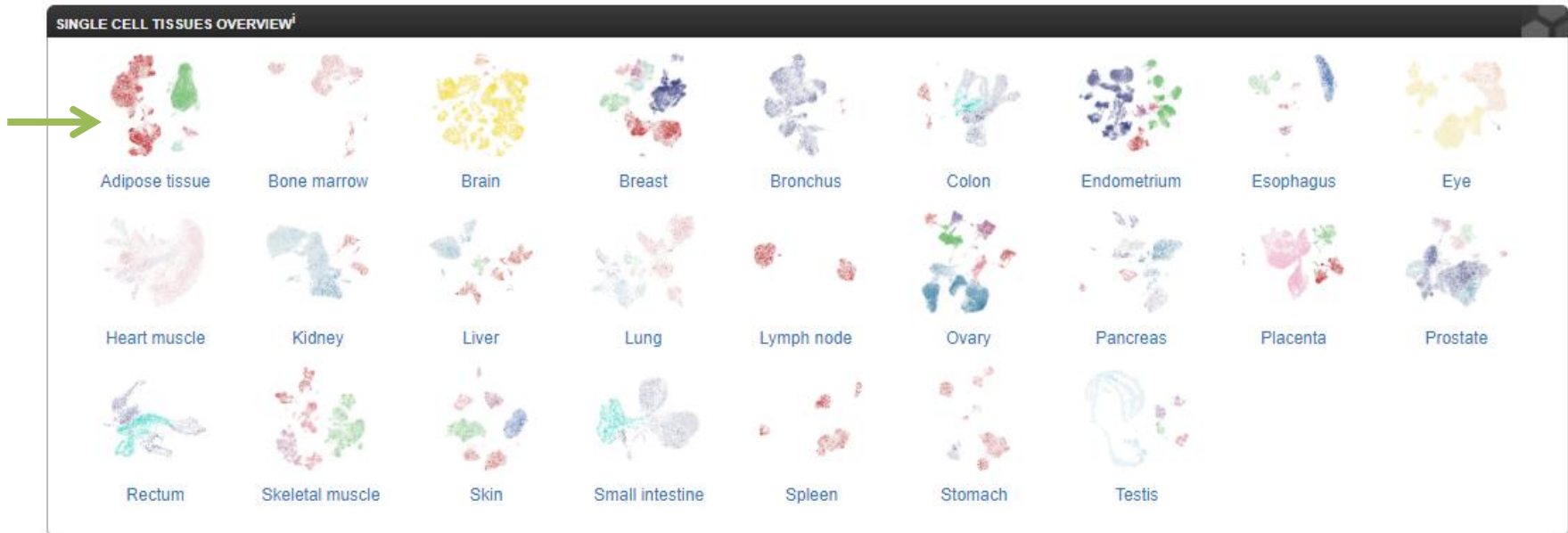
From averages to new revelations



- Expression in **single cells population**



- Expression in **single cells population**



- Expression in **single cells population**

SINGLE CELL TYPE

ADIPOSE TISSUE

GENE/PROTEIN

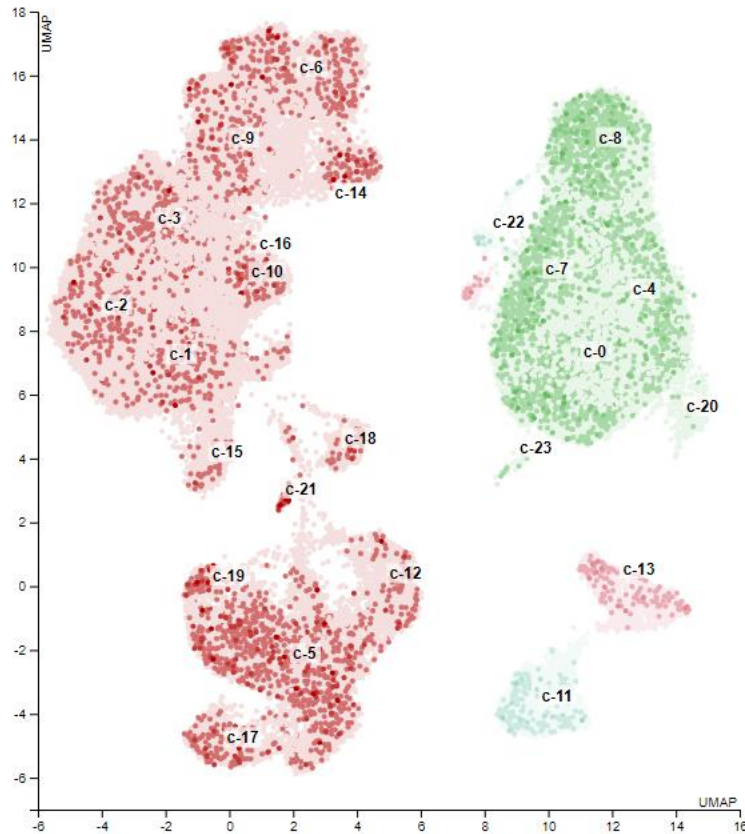
Cell type proteome

- Muscle cells
- Adipocytes
- Mesenchymal cells
- Blood & immune cells

ADIPOSE TISSUE¹

Cell type color Cluster color
Colored according to cell type group

Intensity Interval Hexagon
Scatter plot, all cells color scale - % of max



Cluster	Cell type	Expression (nTPM)
c-0	Fibroblasts	0
c-1	T-cells	10
c-2	T-cells	10
c-3	T-cells	10
c-4	Fibroblasts	0
c-5	Macrophages	10
c-6	Dendritic cells	10
c-7	Fibroblasts	0
c-8	Fibroblasts	0
c-9	T-cells	10
c-10	T-cells	10
c-11	Adipocytes	0
c-12	Macrophages	10
c-13	Smooth muscle cells	10
c-14	Dendritic cells	10
c-15	T-cells	10
c-16	Plasma cells	10
c-17	Macrophages	10
c-18	B-cells	10
c-19	Macrophages	10
c-20	Fibroblasts	0
c-21	Dendritic cells	10
c-22	Adipocytes	0
c-23	Fibroblasts	0

- CELL TYPE MARKERS¹
- Max-norm Z-score
- log(nTPM) z-score
-
- Current gene
- Adipocytes
- B-cells
- Dendritic cells
- Granulocytes
- Macrophages
- Monocytes
- NK-cells
- Plasma cells
- T-cells
- Endothelial cells
- Fibroblasts
- Smooth muscle cells
- TP53
- ADIPOQ
- LIPE
- PLIN1
- CD19
- CR2
- MS4A1
- GZMB
- IL3RA
- CEBPE
- HDC
- MS4A2
- CD163
- CD68
- MARCO
- MRC1
- MSR1
- FCGR3A
- KIR2DL4
- PIM2
- CD3E
- CD4
- CD8A
- FOXP3
- IL17A
- CD34
- PECAM1
- SELE
- SLC2A1
- VWF
- COL3A1
- FBN1
- LUM
- ACTA2
- ACTG2
- CNN1
- MYH11
- Fibroblasts c-0
- T-cells c-1
- T-cells c-2
- T-cells c-3
- Fibroblasts c-4
- Macrophages c-5
- Dendritic cells c-6
- Fibroblasts c-7
- Fibroblasts c-8
- T-cells c-9
- T-cells c-10
- Adipocytes c-11
- Macrophages c-12
- Smooth muscle cells c-13
- Dendritic cells c-14
- T-cells c-15
- Plasma cells c-16
- B-cells c-17
- Macrophages c-18
- Macrophages c-19
- Fibroblasts c-20
- Dendritic cells c-21
- Adipocytes c-22
- Fibroblasts c-23
- 4
- 2
- 0
- 2
- 4

- Physical interactions involving the protein

TP53

SUMMARY

TISSUE

BRAIN

SINGLE CELL

TISSUE CELL

PATHOLOGY

DISEASE

IMMUNE

BLOOD

SUBCELL

CELL LINE

STRUCTURE

INTERACTION

TP53

TP53

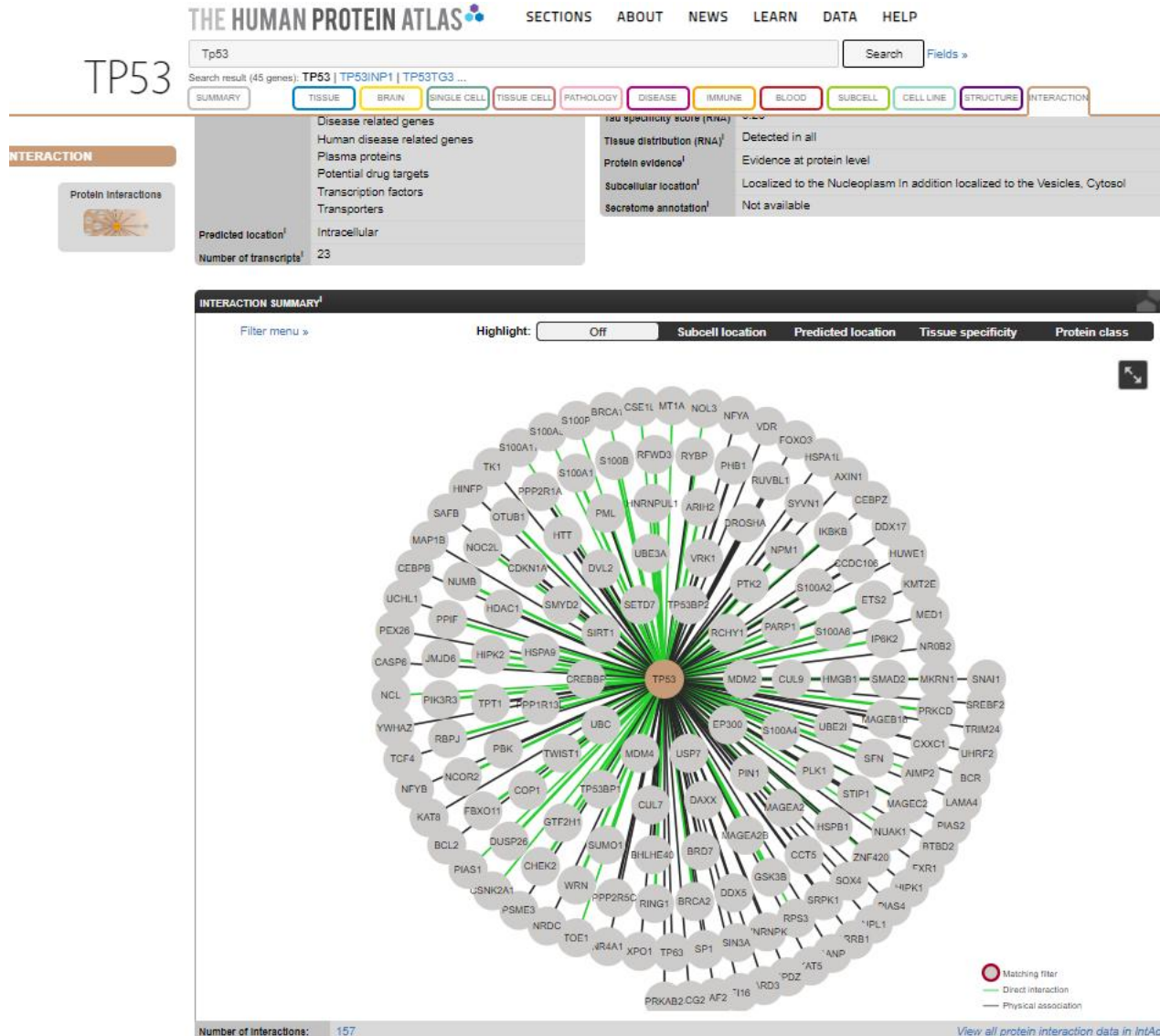
TP53INP1

TP53TG3

TP53 INFORMATION

Protein ⁱ	Tumor protein p53
Gene name ⁱ	TP53 (LFS1, p53)
Protein class ⁱ	Cancer-related genes Disease related genes Human disease related genes Plasma proteins Potential drug targets Transcription factors Transporters
Protein evidence	Evidence at protein level (all genes)
Number of transcripts ⁱ	23
Protein interactions	Interacting with 157 proteins

- **Physical interactions involving the protein**



- **Direct interaction:** Interaction that happens between molecules that are in direct contact with each other. We reserve this annotation for evidence found through methods that use purified molecules, not cell extracts or other complex samples, so we can 100% sure that there are no undetected participants bridging the interaction (e.g. the structure of two co-crystallized, purified proteins)
- **Physical association:** Interaction between molecules within the same physical complex, complex being defined as a group of molecules that are physically in contact at the same point in time. A physical association does not imply a direct contact, even if it refers to an interaction with only two participants. The annotation implies they are together in the same complex, but there could be undetected participants that bridge the interaction, so no proof of directness is given.

Data from IntAct database: <https://www.ebi.ac.uk/intact/documentation/user-guide>